How to Prove: Finite VC Dimension ⇒ Uniform Convergence

A Guided Journey Through Symmetrization

DSAA 3073 · Week 3 Support Material

1. Introduction: The Challenge

Goal: Prove that if $\mathrm{VCdim}(\mathcal{H}) = d < \infty$, then $\mathcal{H}$ has the **uniform convergence** property:

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \to 0 \text{ as } m \to \infty \tag{1}$$

with high probability over random samples $S \sim \mathcal{D}^m$.

Standing Assumptions

Throughout this document, we assume:

1. **Bounded loss:** $\mathrm{loss} : \mathcal{H} \times \mathcal{X} \times \mathcal{Y} \to [0, 1]$ (losses bounded in $[0, 1]$)
2. **i.i.d. sampling:** Training samples $S = \{z_1, ..., z_m\}$ drawn independently from distribution $\mathcal{D}$
3. **Finite VC dimension:** $\mathrm{VCdim}(\mathcal{H}) = d < \infty$

These assumptions are standard and essential for the results that follow.

Why is this hard? The obstacle is that we want to bound:

$$\mathbb{P}\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| > \varepsilon\right] \tag{2}$$

but this involves the **supremum over infinitely many hypotheses**. We can't use a union bound directly!

The solution: Use **symmetrization** to replace the unknown true distribution $\mathcal{D}$ with a "ghost" copy of the training data.

2. The Proof Strategy: A Roadmap

Before diving into details, let's understand the **architecture** of the proof:

Three-Stage Proof Architecture

Stage 1: Symmetrization (Replace unknown with known)

- Replace $R(h)$ (unknown expectation) with $\hat{R}_{S'}(h)$ (empirical average on ghost sample)
- Key tool: Introduce independent "ghost sample" S' of same size
- Result: Reduce to comparing two empirical averages

Stage 2: Concentration via Rademacher Complexity (Exploit finite VC dimension)

- Bound supremum using **Rademacher complexity** $\mathrm{Rad}_m(\mathcal{H})$
- Connect Rademacher complexity to growth function via random signs
- Use Massart's lemma to bound in terms of $|\mathcal{H}|_S|$ (number of dichotomies)

Stage 3: Growth Function Bound (Apply Sauer's Lemma)

- Use Sauer's lemma: $|\mathcal{H}|_S| \le \sum_{i=0}^{d} \binom{m}{i} \le \left(e\frac{m}{d}\right)^d$
- Polynomial growth $\Rightarrow$ logarithmic contribution to sample complexity
- Final bound: $m = O\left(\frac{d}{\varepsilon^2}\right)$ samples suffice

3. Stage 1: Symmetrization—The Ghost Sample Trick

3.1. The Core Intuition

The problem: We want to bound:

$$\mathbb{P}_S\left[\sup_{h\in\mathcal{H}}\left(R(h) - \hat{R}_S(h)\right) > \varepsilon\right] \tag{3}$$

But $R(h) = \mathbb{E}_{z\sim\mathcal{D}}[\text{loss}(h, z)]$ is an expectation over the **unknown** distribution $\mathcal{D}$. We can't compute it directly.

The insight: Replace the unknown expectation $R(h)$ with an empirical average on a **second independent sample**.

3.2. Visualization: Two Parallel Universes

Imagine two parallel experiments:

Experiment 1 (Training Sample S):

$$S = \{z_1, z_2, ..., z_m\} \text{ where } z_i = (x_i, y_i) \sim \mathcal{D} \tag{4}$$

Empirical risk: $\hat{R}_S(h) = \frac{1}{m}\sum_{i=1}^{m} \text{loss}(h, z_i)$

Experiment 2 (Ghost Sample S'):

$$S' = \{z'_1, z'_2, ..., z'_m\} \text{ where } z'_i = (x'_i, y'_i) \sim \mathcal{D} \tag{5}$$

Empirical risk: $\hat{R}_{S'}(h) = \frac{1}{m}\sum_{i=1}^{m} \text{loss}(h, z'_i)$

Key observation: Both $\hat{R}_S(h)$ and $\hat{R}_{S'}(h)$ are unbiased estimates of $R(h)$:

$$\mathbb{E}_S\left[\hat{R}_S(h)\right] = R(h) \quad \text{and} \quad \mathbb{E}_{S'}\left[\hat{R}_{S'}(h)\right] = R(h) \tag{6}$$

3.3. The Symmetrization Lemma

Lemma 1 (Symmetrization):

For any $\varepsilon > 0$:

$$\mathbb{P}_S\left[\sup_{h\in\mathcal{H}}\left(R(h) - \hat{R}_S(h)\right) > \varepsilon\right] \leq 2 \cdot \mathbb{P}_{S,S'}\left[\sup_{h\in\mathcal{H}}\left(\hat{R}_{S'}(h) - \hat{R}_S(h)\right) > \frac{\varepsilon}{2}\right] \tag{7}$$

Proof intuition: The event $R(h) - \hat{R}_S(h) > \varepsilon$ means the true risk exceeds the training risk significantly. If this happens, then with good probability, a second sample S' would also show higher empirical risk than S.

Formal proof: Define the event:

$$A = \left\{S : \sup_{h\in\mathcal{H}}\left(R(h) - \hat{R}_S(h)\right) > \varepsilon\right\} \tag{8}$$

We want to bound $\mathbb{P}[A]$. Suppose A occurs, i.e., there exists $h^* \in \mathcal{H}$ with:

$$R(h^*) - \hat{R}_S(h^*) > \varepsilon \tag{9}$$

Now consider a ghost sample $S' \sim \mathcal{D}^m$ independent of S. We have:

$$\mathbb{E}_{S'}\left[\hat{R}_{S'}(h^*)\right] = R(h^*) \tag{10}$$

By Markov's inequality applied to the deviation:

$$\mathbb{P}_{S'}\left[\hat{R}_{S'}(h^*) - \hat{R}_S(h^*) > \frac{\varepsilon}{2} \mid S\right] \geq \mathbb{P}_{S'}\left[\hat{R}_{S'}(h^*) > R(h^*) - \frac{\varepsilon}{2} \mid S\right] \tag{11}$$

Since $R(h^*) > \hat{R}_S(h^*) + \varepsilon$, we have:

$$R(h^*) - \frac{\varepsilon}{2} > \hat{R}_S(h^*) + \frac{\varepsilon}{2} \tag{12}$$

Therefore:

$$\mathbb{P}_{S'}\left[\sup_{h\in\mathcal{H}}\left(\hat{R}_{S'}(h) - \hat{R}_S(h)\right) > \frac{\varepsilon}{2} \mid S\right] \geq \frac{1}{2} \tag{13}$$

Taking expectation over S:

$$\mathbb{P}_{S,S'}\left[\sup_{h\in\mathcal{H}}\left(\hat{R}_{S'}(h) - \hat{R}_S(h)\right) > \frac{\varepsilon}{2}\right] \geq \frac{1}{2} \cdot \mathbb{P}_S[A] \tag{14}$$

Rearranging gives the result. qed

3.4. Why Does This Help?

Before symmetrization: We had to deal with the unknown $R(h)$.

After symmetrization: We only need to bound:

$$\mathbb{P}_{S,S'}\left[\sup_{h\in\mathcal{H}}|\hat{R}_{S'}(h) - \hat{R}_S(h)| > \varepsilon\right] \tag{15}$$

This is a statement about **two empirical averages on finite samples**—much more tractable!

4. Stage 2: Rademacher Complexity

4.1. The Symmetry Observation

Consider the difference $\hat{R}_{S'}(h) - \hat{R}_S(h)$:

$$\hat{R}_{S'}(h) - \hat{R}_S(h) = \frac{1}{m}\sum_{i=1}^{m}[\mathrm{loss}(h, z_i') - \mathrm{loss}(h, z_i)] \tag{16}$$

Key insight: Since z_i and z_i' are **i.i.d.** draws from $\mathcal{D}$, they are **exchangeable**. The distribution of the difference is the same as:

$$\frac{1}{m}\sum_{i=1}^{m}\sigma_i[\mathrm{loss}(h, z_i') - \mathrm{loss}(h, z_i)] \tag{17}$$

where $\sigma_i \in \{\pm 1\}$ are **independent Rademacher random variables** (uniform on $\{-1, +1\}$).

4.2. Visualization: Random Signs as Symmetry Test

Think of Rademacher variables as **coin flips** deciding whether to swap z_i and z_i':

If $\sigma_i = +1$: keep (z_i, z_i') in original order
If $\sigma_i = -1$: swap to (z_i', z_i)

Since the joint distribution is symmetric, the **supremum of the difference** has the same distribution as:

$$\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i[\mathrm{loss}(h, z_i') - \mathrm{loss}(h, z_i)] \tag{18}$$

4.3. Rademacher Complexity Definition

Definition (Rademacher Complexity):

Given a sample $S = \{z_1, ..., z_m\}$ and hypothesis class $\mathcal{H}$:

Empirical Rademacher complexity (random over σ only):

$$\mathrm{Rad}_S(\mathcal{H}) = \mathbb{E}_\sigma\left[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i \ \mathrm{loss}(h, z_i)\right] \tag{19}$$

where $\sigma = (\sigma_1, ..., \sigma_m)$ are i.i.d. Rademacher variables (uniform on $\{-1, +1\}$).

Expected Rademacher complexity (expected over both S and σ):

$$\mathrm{Rad}_m(\mathcal{H}) = \mathbb{E}_S[\mathrm{Rad}_S(\mathcal{H})] = \mathbb{E}_{S,\sigma}\left[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i \ \mathrm{loss}(h, z_i)\right] \tag{20}$$

Important distinction: $\mathrm{Rad}_S(\mathcal{H})$ depends on the specific sample S, while $\mathrm{Rad}_m(\mathcal{H})$ is a fixed number (the expectation over random samples).

Interpretation: Rademacher complexity measures **how well $\mathcal{H}$ can fit random noise**.

Why "correlation"? The term comes from viewing $\frac{1}{m}\sum_{i=1}^{m} \sigma_i \operatorname{loss}(h, z_i)$ as an **empirical covariance** between:

- The loss vector: $(\operatorname{loss}(h, z_1), ..., \operatorname{loss}(h, z_m))$
- The random sign vector: $(\sigma_1, ..., \sigma_m)$

Since $\mathbb{E}[\sigma_i] = 0$ and $\operatorname{loss}(h, z_i) \in [0, 1]$, this sum measures how much the losses **align** with random signs. High values mean $\mathcal{H}$ can find patterns even in pure noise—a sign of overfitting capacity.

More precisely: for centered variables X and Y with unit variance, correlation is $\mathbb{E}[XY]$. Here, σ_i are centered (mean 0), and we're computing $\mathbb{E}_\sigma[\sum \sigma_i \operatorname{loss}(h, z_i)]$, which is the empirical correlation scaled by m.

If $\mathcal{H}$ can achieve large positive values of this sum, it has high complexity.

4.4. Connecting to Uniform Convergence

Using Rademacher complexity, we can bound:

Lemma 2 (Uniform Convergence Bound via Rademacher):

With probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$:

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \leq 2 \operatorname{Rad}_m(\mathcal{H}) + \sqrt{\frac{\log(\frac{2}{\delta})}{2m}} \tag{21}$$

4.4.1. Complete Proof of Lemma 2

Step 1: Apply symmetrization (Lemma 1)

From Lemma 1, we have:

$$\mathbb{P}_S\left[\sup_{h \in \mathcal{H}}\left(R(h) - \hat{R}_S(h)\right) > \varepsilon\right] \leq 2\mathbb{P}_{S,S'}\left[\sup_{h \in \mathcal{H}}\left(\hat{R}_{S'}(h) - \hat{R}_S(h)\right) > \frac{\varepsilon}{2}\right] \tag{22}$$

Step 2: Use symmetry to introduce Rademacher variables

Since S and S' are i.i.d. samples from $\mathcal{D}$, the pairs (z_i, z_i') are exchangeable. Therefore:

$$\hat{R}_{S'}(h) - \hat{R}_S(h) = \frac{1}{m}\sum_{i=1}^{m}[\operatorname{loss}(h, z_i') - \operatorname{loss}(h, z_i)] \tag{23}$$

has the same distribution as:

$$\frac{1}{m}\sum_{i=1}^{m}\sigma_i[\operatorname{loss}(h, z_i') - \operatorname{loss}(h, z_i)] \tag{24}$$

where $\sigma_i \in \{\pm 1\}$ are independent Rademacher random variables.

Why? Flipping the sign is equivalent to swapping z_i and z_i', which doesn't change the joint distribution since they're i.i.d.

Step 3: Bound expectation using Rademacher complexity

Taking expectation over S, S', σ:

$$\mathbb{E}_{S,S',\sigma}\left[\sup_{h \in \mathcal{H}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i[\operatorname{loss}(h, z_i') - \operatorname{loss}(h, z_i)]\right] \tag{25}$$

By triangle inequality for the supremum:

$$\leq \mathbb{E}_{S,S',\sigma}\left[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i \ \mathrm{loss}(h, z_i')\right] + \mathbb{E}_{S,S',\sigma}\left[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^{m}(-\sigma_i) \ \mathrm{loss}(h, z_i)\right] \tag{26}$$

Since σ_i and $-\sigma_i$ have the same distribution, and S and S' are identically distributed:

$$= 2\mathbb{E}_{S,\sigma}\left[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i \ \mathrm{loss}(h, z_i)\right] = 2\,\mathrm{Rad}_m(\mathcal{H}) \tag{27}$$

Step 4: Apply concentration inequality

The function $\varphi(S, S') = \sup_{h\in\mathcal{H}}\left(\hat{R}_{S'}(h) - \hat{R}_S(h)\right)$ satisfies the **bounded differences property**: Changing one sample point z_i can change φ by at most $\frac{2}{m}$ (since losses are in $[0, 1]$).

By McDiarmid's inequality, with probability at least $1 - \delta$:

$$\varphi(S, S') \leq \mathbb{E}[\varphi(S, S')] + \sqrt{\frac{\log(\frac{1}{\delta})}{2 \cdot (2m)}} \tag{28}$$

where the factor $2m$ comes from m points in S plus m points in S'.

Step 5: Combine to get the final bound

From Steps 3 and 4:

$$\mathbb{P}_{S,S'}\left[\sup_{h\in\mathcal{H}}\left(\hat{R}_{S'}(h) - \hat{R}_S(h)\right) > 2\,\mathrm{Rad}_m(\mathcal{H}) + \sqrt{\frac{\log(\frac{1}{\delta})}{2m}}\right] \leq \delta \tag{29}$$

Plugging into Step 1 with $\varepsilon = 4\,\mathrm{Rad}_m(\mathcal{H}) + 2\sqrt{\frac{\log(\frac{1}{\delta})}{2m}}$:

$$\mathbb{P}_{S}\left[\sup_{h\in\mathcal{H}}\left(R(h) - \hat{R}_S(h)\right) > 2\,\mathrm{Rad}_m(\mathcal{H}) + \sqrt{\frac{\log(\frac{2}{\delta})}{2m}}\right] \leq 2\delta \tag{30}$$

Step 6: Two-sided bound via symmetry

The above gives a **one-sided** bound on $R(h) - \hat{R}_S(h)$ (positive deviations). To get the two-sided bound $|R(h) - \hat{R}_S(h)|$, we apply the same argument to $\hat{R}_S(h) - R(h)$ (negative deviations).

By **union bound** over both directions:

$$\mathbb{P}_{S}\left[\sup_{h\in\mathcal{H}}|R(h) - \hat{R}_S(h)| > 2\,\mathrm{Rad}_m(\mathcal{H}) + \sqrt{\frac{\log(\frac{2}{\delta})}{2m}}\right] \leq \delta \tag{31}$$

Note: The factor of 2 in $\log(\frac{2}{\delta})$ comes from applying union bound to control both $\pm$ deviations simultaneously. This is where we "pay" for measuring absolute deviation rather than just one-sided excess risk.

This completes the proof. ■

Key insight: The proof bridges three ideas:
1. **Symmetrization** replaces the unknown $R(h)$ with empirical $\hat{R}_{S'}(h)$
2. **Rademacher variables** encode the symmetry and measure fitting to noise
3. **Concentration** (McDiarmid) gives high-probability bounds from the expectation

5. Stage 3: Bounding Rademacher Complexity via Growth Function

5.1. The Finite Dichotomy Trick

Key observation: Although $\mathcal{H}$ may be infinite, on any fixed sample $S = \{z_1, ..., z_m\}$, there are only **finitely many distinct behaviors**.

Define the **restriction** of $\mathcal{H}$ to S:

$$\mathcal{H}|_S = \{(\text{loss}(h, z_1), ..., \text{loss}(h, z_m)) : h \in \mathcal{H}\} \quad (32)$$

For binary classification with 0-1 loss:

$$\mathcal{H}|_S = \left\{\left(\mathbb{1}_{h(x_1)\neq y_1}, ..., \mathbb{1}_{h(x_m)\neq y_m}\right) : h \in \mathcal{H}\right\} \quad (33)$$

The cardinality $|\mathcal{H}|_S|$ is the number of **distinct dichotomies** on S.

5.2. Massart's Lemma

Lemma 3 (Massart's Lemma):

Let $A \subseteq \{-1, +1\}^m$ be a finite set of vectors. Then:

$$\mathbb{E}_\sigma\left[\sup_{a\in A} \frac{1}{m}\sum_{i=1}^{m} \sigma_i a_i\right] \leq \frac{\sqrt{\log|A|}}{\sqrt{m}} \quad (34)$$

Proof intuition: The supremum over A introduces a $\log|A|$ factor (similar to union bound), but the random signs provide concentration.

Rigorous proof: Massart's lemma uses the **contraction lemma** for Rademacher variables, which states that replacing variables with their linear combinations cannot increase Rademacher complexity. For a complete proof, see Mohri et al. (Foundations of Machine Learning, 2nd ed.), Theorem 3.7, or Ledoux & Talagrand (1991). The key technique is showing the expected supremum is controlled by $\sqrt{\log|A\frac{|}{m}}$ via concentration of suprema of sub-Gaussian processes.

5.3. Connecting to VC Dimension

For binary classification, the loss vectors form a finite set:

$$A = \{(\text{loss}(h, z_1), ..., \text{loss}(h, z_m)) : h \in \mathcal{H}\} \subseteq \{0, 1\}^m \quad (35)$$

with $|A| = |\mathcal{H}|_S|$.

By Massart's lemma:

$$\text{Rad}_S(\mathcal{H}) \leq \sqrt{\frac{\log|\mathcal{H}|_S|}{m}} \quad (36)$$

Now use **Sauer's lemma**:

Lemma 4 (Sauer's Lemma):

If $\text{VCdim}(\mathcal{H}) = d < \infty$, then for all m:

$$|\mathcal{H}|_S| \leq \sum_{i=0}^{d} \binom{m}{i} \leq \left(\frac{em}{d}\right)^d \quad (37)$$

Therefore:

$$\mathrm{Rad}_m(\mathcal{H}) \leq \sqrt{\frac{d \log\left(e\frac{m}{d}\right)}{m}} = O\left(\sqrt{\frac{d \log m}{m}}\right) \tag{38}$$

6. Stage 4: Putting It All Together

6.1. The Complete Bound

Combining all pieces:

Theorem (Finite VC Dimension ⇒ Uniform Convergence):

Let $\mathcal{H}$ be a hypothesis class with $\mathrm{VCdim}(\mathcal{H}) = d < \infty$. Then with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$:

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \leq O\left(\sqrt{\frac{d \log m + \log\left(\frac{1}{\delta}\right)}{m}}\right) \tag{39}$$

In particular, for $m = O\left(\frac{d + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right)$, we have uniform convergence with ε accuracy.

6.2. Proof Summary and Key Insights

What we proved: $\mathrm{VCdim}(\mathcal{H}) = d < \infty \Rightarrow$ Uniform convergence with $m = O\left(\frac{d}{\varepsilon^2}\right)$ samples.

The proof architecture:

1. **Symmetrization (Lemma 1):** Replace $R(h)$ with $\hat{R}_{S'}(h)$ using ghost sample
   - Key insight: Reduces problem to comparing two empirical averages
   - Cost: Factor of 2 loss in error parameter
2. **Rademacher complexity (Lemma 2):** Use symmetry of (S, S') to introduce random signs
   - Key insight: Measures how well $\mathcal{H}$ can correlate with random noise
   - Method: Triangle inequality splits the bound, giving $2\,\mathrm{Rad}_m(\mathcal{H})$
   - Concentration: McDiarmid's inequality provides high-probability bound
3. **Massart's lemma (Lemma 3):** Bound Rademacher via number of dichotomies
   - Key insight: Finite dichotomies ⇒ logarithmic contribution: $\sqrt{\frac{\log|\mathcal{H}|_S|}{m}}$
   - Method: Supremum over finite set introduces $\log|A|$ factor with concentration
4. **Sauer's lemma (Lemma 4):** Bound dichotomies via VC dimension
   - Key insight: $\mathrm{VCdim} = d \Rightarrow |\mathcal{H}|_S| \leq \left(e\frac{m}{d}\right)^d$
   - Result: Polynomial growth ⇒ logarithmic contribution to bound
5. **Final result:** Sample complexity $m = O\left(\frac{d}{\varepsilon^2}\right)$

Why the proof works:
- Ghost samples let us avoid reasoning about unknown distribution $\mathcal{D}$
- Random signs encode symmetry and enable concentration inequalities
- Finite VC dimension ⇒ polynomial growth ⇒ logarithmic sample complexity
- The composition bridges combinatorics (VC dimension) and statistics (sample complexity)

Core intuition: A hypothesis class with finite VC dimension has **limited expressiveness** on finite samples. This prevents overfitting and ensures empirical performance predicts true performance.

Optimality of the bounds: The sample complexity $m = O\left(\frac{d}{\varepsilon^2}\right)$ is essentially tight. Matching lower bounds due to Ehrenfeucht et al. [1989] and others show that $\Omega\left(\frac{d}{\varepsilon^2}\right)$ samples are **necessary** in the worst case for agnostic learning. Thus, our upper bound cannot be significantly improved in general (though tighter constants and data-dependent bounds are possible for specific classes).

Historical note: These foundational results are due to:

- Vapnik & Chervonenkis [1971]: VC dimension and growth functions
- Sauer [1972], Shelah [1972]: Independent proofs of the growth function bound (Sauer's lemma)
- Massart [2000]: The finite-class Rademacher bound (Massart's lemma)
- Koltchinskii [2001]: Rademacher complexity in statistical learning theory

7. Visual Mental Models

7.1. The Ghost Sample as a Mirror

Think of S' as a **mirror** of S:

Original sample $S \Leftrightarrow$ Ghost sample S'
Both drawn from same $\mathcal{D}$
If h fits S much better than S', suspect overfitting

The symmetrization trick says: **a hypothesis that generalizes well should perform similarly on both samples**.

7.2. Rademacher Variables as Stress Test

Random signs σ_i act as a **stress test**:

- If $\mathcal{H}$ can achieve high correlation with random ± 1 labels, it's too flexible
- Low Rademacher complexity means $\mathcal{H}$ **resists fitting noise**

7.3. Growth Function as Effective Size

Even though $|\mathcal{H}| = \infty$, on any m points, only $|\mathcal{H}|_S| \leq \left(\frac{m}{d}\right)^d$ distinct behaviors exist:

Infinite hypotheses $\Rightarrow$ Polynomial effective behaviors
$\log|\mathcal{H}|_S| \approx d \log m \Rightarrow$ Acts like $d \log m$ "effective parameters"

8. Common Confusions and Clarifications

8.1. Why Do We Need Ghost Samples?

Question: Can't we just use Hoeffding's inequality on $R(h) - \hat{R}_S(h)$ for each h?

Answer: Hoeffding works for a **single** fixed hypothesis. But we need **uniform** convergence over **all** $h \in \mathcal{H}$. The ghost sample trick lets us handle the supremum.

8.2. Why Random Signs?

Question: Why introduce Rademacher variables σ_i?

Answer: They encode **symmetry**. Since (z_i, z_i') are exchangeable, flipping their order randomly doesn't change the distribution. This lets us convert the problem into **fitting random noise**, which is easier to analyze.

8.3. Why Does Sauer's Lemma Matter?

Question: Why is the growth function bound important?

Answer: It's the **bridge** from VC dimension (a combinatorial property) to sample complexity (a statistical quantity). Without Sauer's lemma, we couldn't prove that $\text{VCdim} < \infty$ implies learnability.

8.4. The ε^2 vs. ε Mystery

Question: Why does agnostic learning need $m = O\left(\frac{1}{\varepsilon^2}\right)$ while realizable needs $m = O\left(\frac{1}{\varepsilon}\right)$?

Answer:

- **Realizable:** We only need to **detect** bad hypotheses ($R(h) > \varepsilon$). This is a **one-sided** event.
- **Agnostic:** We need to **estimate** all risks accurately (within ε). This requires **two-sided** concentration, which costs ε^2.

9. Exercises for Understanding

9.1. Exercise 1: Symmetrization Practice

Show that if X and Y are i.i.d. with $\mathbb{E}[X] = \mu$:

$$\mathbb{P}[X - \mu > \varepsilon] \leq 2\mathbb{P}[X - Y > \varepsilon] \tag{40}$$

9.2. Exercise 2: Rademacher Computation

Compute the Rademacher complexity for: a) $\mathcal{H} = \{h_1, h_2, ..., h_N\}$ finite class b) $\mathcal{H} =$ intervals on $\mathbb{R}$ with $m = 3$ points

9.3. Exercise 3: Growth Function Bound

Verify Sauer's lemma for $\text{VCdim} = 2$ (intervals) and $m = 3$ points by explicitly enumerating all dichotomies.

9.4. Exercise 4: The Union Bound Comparison

Compare the uniform convergence bound:

$$m = O\left(\frac{d \log m}{\varepsilon^2}\right) \tag{41}$$

with the naive union bound (if $|\mathcal{H}| = N < \infty$):

$$m = O\left(\frac{\log N}{\varepsilon^2}\right) \tag{42}$$

For linear classifiers in $\mathbb{R}^{10}$, which is tighter?

10. Advanced Topics and Extensions

10.1. Rademacher Complexity for Regression

For regression with squared loss, the same framework applies but:

- Losses are in $[0, 1]$ instead of $\{0, 1\}$
- Rademacher complexity measures **correlation** with random ± 1 weights
- Massart's lemma still applies

10.2. Chaining and Localization

More sophisticated bounds use **local Rademacher complexity**:

$$\mathrm{Rad}_m(\mathcal{H}, r) = \mathbb{E}_S\left[\mathbb{E}_\sigma\left[\sup_{h:R(h)\leq r} \frac{1}{m}\sum_{i=1}^{m} \sigma_i \ \mathrm{loss}(h, z_i)\right]\right] \tag{43}$$

This exploits the fact that low-risk hypotheses have lower complexity.

10.3. Connection to Margin Theory (Boosting)

In boosting, we care about **margin** $y \cdot f(x)$ instead of just correctness. Rademacher complexity of **margin-based** loss classes appears in boosting generalization bounds.

10.4. Algorithmic Stability as Alternative

An alternative proof route uses **algorithmic stability**:

- Measure how much the output changes when one training example is perturbed
- Stable algorithms generalize well
- ERM on hypothesis classes with finite VC dimension is stable

11. The Fundamental Lesson

Why Restrictions Enable Learning

Finite VC dimension $\Leftrightarrow$ polynomial growth $\Leftrightarrow$ logarithmic sample complexity

Restrictions on $\mathcal{H}$ are not limitations—they're **what makes learning possible**.